

Digits Identifier Circuit for Euler's Number

PC/CP220 Project Phase II

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October 31st 2024

Truth Table:

A3	A2	A1	A0	Decimal place	e digit	O0	O1	O2	O3	O4	O5	O6	O7	O8	O9
0	0	0	0	0	2	1	0	0	0	0	0	0	0	0	0
0	0	0	1	1	7	0	0	0	0	0	0	0	1	0	0
0	0	1	0	2	1	0	1	0	0	0	0	0	0	0	0
0	0	1	1	3	8	0	0	0	0	0	0	0	0	1	0
0	1	0	0	4	2	1	0	0	0	0	0	0	0	0	0
0	1	0	1	5	8	0	0	0	0	0	0	0	0	1	0
0	1	1	0	6	1	0	1	0	0	0	0	0	0	0	0
0	1	1	1	7	8	0	0	0	0	0	0	0	0	1	0
1	0	0	0	8	2	1	0	0	0	0	0	0	0	0	0
1	0	0	1	9	8	0	00	0	0	0	0	0	0	1	0

Logic Equation:

1. O_0 (Digit 2 at positions 0, 4, and 8)

$$O_0 = A_3' \cdot A_2' \cdot A_1' \cdot A_0' + A_3' \cdot A_2 \cdot A_1' \cdot A_0' + A_3 \cdot A_2' \cdot A_1' \cdot A_0'$$

2. O_1 (Digit 1 at positions 2 and 6):

$$O_1 = A_3' \cdot A_2' \cdot A_1 \cdot A_0' + A_3' \cdot A_2 \cdot A_1 \cdot A_0'$$

3. O_2 (No positions correspond to Digit 2 specifically, so O_2 is always 0):

$$O_2 = 0$$

4. O_3 (No positions correspond to Digit 3 specifically, so O_3 is always 0):

$$O_3 = 0$$

5. O_4 (No positions correspond to Digit 4 specifically, so O_4 is always 0):

$$O_4 = 0$$

6. O_5 (No positions correspond to Digit 5 specifically, so O_5 is always 0):

$$O_5 = 0$$

7. O_6 (No positions correspond to Digit 6 specifically, so O_6 is always 0):

$$O_6 = 0$$

8. O_7 (Digit 7 at position 1):

$$O_7 = A_3' \cdot A_2' \cdot A_1' \cdot A_0$$

9. O_8 (Digit 8 at positions 3, 5, 7, and 9):

$$O_8 = A_3' \cdot A_2' \cdot A_1 \cdot A_0 + A_3' \cdot A_2 \cdot A_1' \cdot A_0 + A_3' \cdot A_2 \cdot A_1 \cdot A_0 + A_3 \cdot A_2' \cdot A_1' \cdot A_0$$

10. O_9 (No positions correspond to Digit 5 specifically, so O_9 is always 0):

$$O_9 = 0$$

Testing Equation:

Input:

t1: not(a3) and not(a2) and not(a1) and not(a0) or not(a3) and a2 and not(a1) and not(a0) or a3 and not(a2) and not(a1) and not(a0);
t2: not(a3) and not(a2) and a1 and not(a0) or not(a3) and a2 and a1 and not(a0);
t3: not(a3) and not(a2) and not(a1) and a0;
t4: not(a3) and not(a2) and a1 and a0 or a3 and not(a2) and not(a1) and a0 or not(a3) and a2 and a1 and a0 or a3 and not(a2) and not(a1) and a0;

euler: t1 or t2 or t3 or t4;

```
euler,a0=false,a1=false,a2=false,a3=false;  
euler,a0=true,a1=false,a2=false,a3=false;  
euler,a0=false,a1=true,a2=false,a3=false;  
euler,a0=true,a1=true,a2=false,a3=false;  
euler,a0=false,a1=false,a2=true,a3=false;  
euler,a0=true,a1=false,a2=true,a3=false;  
euler,a0=false,a1=true,a2=true,a3=false;  
euler,a0=true,a1=true,a2=true,a3=false;  
euler,a0=false,a1=false,a2=false,a3=true;  
euler,a0=true,a1=false,a2=false,a3=true;
```

Output:

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¬a3 ∧ ¬a2 ∧ ¬a1 ∧ ¬a0 ∨ ¬a3 ∧ a2 ∧ ¬a1 ∧ ¬a0 ∨ a3 ∧ ¬a2 ∧ ¬a1 ∧ ¬a0  
¬a3 ∧ ¬a2 ∧ a1 ∧ ¬a0 ∨ ¬a3 ∧ a2 ∧ a1 ∧ ¬a0  
¬a3 ∧ ¬a2 ∧ ¬a1 ∧ a0  
¬a3 ∧ ¬a2 ∧ a1 ∧ a0 ∨ a3 ∧ ¬a2 ∧ ¬a1 ∧ a0 ∨ ¬a3 ∧ a2 ∧ a1 ∧ a0 ∨ a3 ∧ ¬a2 ∧ ¬a1 ∧ a0  
¬a3 ∧ ¬a2 ∧ ¬a1 ∧ a0 ∨ a3 ∧ a2 ∧ ¬a1 ∧ a0 ∨ ¬a3 ∧ a2 ∧ a1 ∧ a0 ∨ a3 ∧ ¬a2 ∧ ¬a1 ∧ a0 ∨ a3 ∧ a2 ∧ a1 ∧ a0 ∨ a3 ∧ ¬a2 ∧ a1 ∧ a0  
true  
true  
true  
true  
true  
false  
true  
true  
¬A0  
true
```

Summary:

The Digit Identifier Circuit decodes certain decimal figures of Euler's number e. In other words, for operation we are using the 4-bit binary input, wherein each input corresponds to the decimal value between 0 and 9. The circuit outputs each representative equivalent binary of the compared digit of e included, for example, 2 as the output corresponding to input 0000 and 7 as the output to input 0001. If an input is greater or equal to 10 in decimal or non-binary all outputs are turned off and set to 0 to indicate an error. Each output is on only when the decimal place for the involved number matches the desired one, in order to make each valid input to correspond to a one-hot encoding. This circuit design utilizes logic equations that govern the outputs in similar structures laboratory reviewed to ensure accuracy and range compliance.